Consultant Profile

Dieter Eickstädt

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Dieter Eickstädt

Personal data

Surname	Eickstädt
Name	Dieter
Year of Birth	1972
Nationality	deutsch
Foreign Languages	Englisch
IT experience since	1994
E-Mail	eickstaedt@deicon.de
Phone	06128 / 97 90 710
Mobile	0172 / 613 86 72

Education / Certifications

- Diplom Informatiker (FH), Hochschule Rhein Main Wiesbaden.

Business Focus

- Investment Banking
- Risk Management Systems / Calculation Engines
- Initial Margin according to Standard Initial Margin Model (SIMM)™
- Monte Carlo Simulation, VaR

Technical Focus

- Clod Computing (**Kubernetes**, Docker, Azure, AWS)
- High Performance Computing / High Volume Calculations
- Distributed Systems (J2EE, **OSGI**)
- Cross Platform App Development (Android, IOS, Web, Mac, Windows) using Flutter
- Test automation (**Keyword based acceptance testing**)
- **Microservices**
- **Distributed Data Management**
- **Java**, **Flutter**, **Dart**, **GO**, Typescript, C++

Summary

Since the first releases of Java, I have realised extensive projects with the Java language. From working as a permanent ORACLE consultant, to developing project management software (WebProject Inc., San Mateo CA), to various projects as an IT consultant.

The technical focus of my work is in the banking and investment banking environment. Here I have been involved both as a developer and as a sub-project manager in the realisation of complex online trading systems for fixed-interest securities (bonds) and options on foreign exchange (OTC). In recent years, I have also been increasingly involved with highly available and scalable systems in the Java J2EE environment and realised them in concrete projects in risk management (investment bank).

Several highly available and computationally intensive clusters for VaR, exposure and historical simulation calculations were created for a German tier-1 investment bank. These are used for both market and counterparty default risk.

Due to the sometimes high mathematical complexity of the implemented systems, I have placed a great deal of focus on the automation of the technical acceptance tests in the projects. I am constantly developing the resulting framework and actively use it in my customer projects. In the last 10 years, I have mostly taken on the role of technical lead in the projects I have been involved in, and have thus been able to make a significant contribution to the architectures that have emerged.

In the early 2000s, I published my experiences with free software (open source) used in projects as an author in two books published by Markt und Technik Verlag.

Java mit Open Source Werkzeugen	Markt und Technik Verlag	ISBN 3827264626
J2EE mit Struts und Co.	Markt und Technik Verlag	ISBN 3827266907

I like to take the responsibility to make the projects I am working in a success. I have experience in leading technical teams as well as the motivation of the staff - even in critical project environments. I am also used to presenting and discussing project results at senior management level.

In addition, I have successfully implemented and used agile project management methods such as Scrum as a developer and as a Scrum Master at the client.

Technologies

The following list gives an overview of special knowledge that I have acquired in the course of my projects. Special focal points are shown in bold. .General

Operating Systems	**Linux**, **Windows 7/10**
Office	**Excel**,**Word**, PowerPoint, Visio
Cloud Services	**AWS**, **Azure**, Google App Engine, Heroku
	Docker, **DockerCompose**, Kubernetes, Helm, VMWare, ESXi
Others	Elasticsearch, Kibana

Architecture / Design

CASE Tools	**SAP Powerdesigner**,**Plantuml**
Applicationservers	**Karaf OSGI**
Methods	**Scrum**, **OOA/OOD**,

Development

Programming languages	**Java**, **Dart**, C++, **SQL**, **GO**, **Typescript**, PL/SQL, Python
Frameworks	**Flutter** **Hibernate**, **Spring**, **Flutter Android/iOS GUICE**, **Camel**, **Swagger**, Hadoop, Gigspaces
IDE	**IntelliJ IDEA**, **Visual Studio Code**, Eclipse
Build Systems	**Maven**, **Gradle**, **NPM**, CMake, Make, ANT
Versioncontrol	**GIT**, **Git Flow**, Mercurial, SVN
Databases	**PostgreSQL**, **SQLite**, MongoDB, Oracle, Sybase, MySQL,**Couchbase**
Messaging Middleware	AMQP, Azure Eventbus, Kafka

Selected projects until 2010

The following overview of my previous projects is intended to give an impression of the technical and professional focus of my career. Since the beginning of my career, I have worked exclusively as an external consultant on various projects, with the exception of a one-year permanent position as an ORACLE consultant.

Since the relevance of the projects listed here decreases over time, only the last few years are presented in detail.

Cross-company sales platform for financial products

Lead Developer / Member of the overall architecture board, DVAG GmbH

Since August 2022

DVAG offers its clients a wide range of financial services through its own network of over 15,000 advisors. The financial situation as well as the goals and wishes of the clients are recorded by the advisors. This is followed by a selection of DVAG products from the insurance and investment portfolio of selected partner companies.

Since DVAG acts as an intermediary for financial products of numerous partner companies, a central advisory platform is created. Via this web-based platform DVAG's financial advisors are able to create offers from the various product partners and conclude contracts based on them.

The project consists of different sub-projects, which are partly implemented at the product partners and partly within DVAG itself. In my role as the lead developer, I am overseeing 7 individual projects with different technologies within DVAG and moderate the cross-company architecture board.

The goal is to create a platform that offers both the most generic view possible of the product portfolio of DVAG and its partner companies, as well as to create a modular and expandable architecture into which new product partners can be easily integrated in the future.

Herausforderungen der Plattform

- Skalierbarkeit der Plattform (Offene Architektur, Cloud Native)
- Abgestimmte Workflows und Schnittstellen
- Integration unterschiedlicher Produktpartner zeitgleich in CI CD
- Integration unterschiedlicher Cloud Architekturen der beteiligten Teilprojekte (Azure, AWS, On Premise)
- Durchsetzen und Überwachen der Platformweiten Architekturvorgaben und Entwicklungsrichtlinien
- Rechtsichere Vertragsabschlüsse, die den Anforderungen der Produktparntner entsprechen
- Hohe sicherheits- und datenschutzrechtliche Anforderungen an die Implementierungen

My tasks in the project:

- Technical Lead Developer (Kotlin Spring Boot Services, Micro Frontends, Web Komponenten, IOS App, Windows App, Backoffice Connections)
- Member of Architecture Board
- Building and leading the technical teams in the different project phases
- Discuss the technical details and solution designs with the teams
- Responsible for internal tools to support the standardization of development processes (e.g. Code Generator, Support Libs)
- Moderation of the coordination processes with regard to common interfaces

Technologies used in the project:

Technology / Frameworks	Tools	Databases	OS
Kotlin, **TypeSript**, **Azure**	**Docker**, **Kubernetes**, **AZURE**, ServiceBus Enterprise Messaging	CosmosDB, PostgreSQL, MongoDB	Linux

Enterprise Content Management Platform / Document Workflow

Solution Designer, Enterprise Architect, Strateco GmbH Bad Homburg

October 2019 - August 2022

Strateco GmbH supports its client, one of the top 5 Auditing companies, in the implementation of complex M&A projects in the course of which access to a large number of documents is necessary. These documents are to be made available to the participants on a central platform, taking into account complex authorisation concepts.

During the offer phase, flexibly configurable workflows are carried out on individual documents.

The aim of the PoC implementation was to demonstrate the feasibility of the envisaged platform for Strateco's end customer. The PoC was presented to the end customer's committees and successfully transferred into an implementation project.

The application is productive is actively used in high ticket M&A projects. A further goal is now to offer the platform as SaaS and to address a broader customer base.

Challenges of the platform

- Scalability of the platform (hexagonal architecture, cloud native)
- User and role concept
- Flexible design of the platform in order to be able to cover upcoming customer requirements

My tasks in the project::

- Requirements analysis (workshops)
- Enterprise Solution Design
- Product evaluation, market overview
- Implementation of central content repository (document storage)
- Integration of a workflow engine (Camunda)

Technologies used in the project:

Technology / Frameworks	Tools	Databases	OS
Java, TypeSript, **Nuxeo**, MongoDB, Kafka, Elasticsearch, **Camel**, AMQP	Docker, **Kubernetes**, AZURE, ServiceBus Enterprise Messaging	PostgreSQL, MongoDB	Linux

Migration of all host systems to Java / Linux

Performance Analyst, Framework Reviewer, ISTA GmbH, Essen

Mai 2019 bis Oktober 2019

ISTA GmbH offers its customers electronic services for the settlement of service charges for rented properties. ISTA sees itself as a one-stop shop provider, ideally offering all services related to the collection and billing of ancillary costs from a single source. From the energy measuring devices, the recording of the measurement data, the issuing of energy performance certificates to the complete preparation and dispatch of the service charge statement.

All data and processes are currently located on IBM mainframes and were ported to a modern distributed system architecture in the course of a migration project lasting several years. In the process, the data and also the existing programmes (COBOL, VSE) were largely ported automatically to Java, Oracle and Unix by the software of an external service provider.

The aim of my assignment was to support the project in achieving the non-functional performance characteristics. Specifically, the sources and framework components provided by the service provider Anubex had to be analysed for runtime problems and suggestions for their optimisation had to be delivered.

In addition, measures were considered that would lead to a sensible modernisation of the architecture with a view to the future operation and further development of the software.

As a result, the runtime of the batch system was reduced from just over 30 hours to the required <10 hours. This was achieved through a variety of measures.

Among others:

- Optimisation of memory management in the framework
- OptimisationoftheRDBMSaccesslayer(JAVA-Oracle)
- Parallelisation of individual sub-jobs
- Optimisation of data management (triggers, indices, where-clauses)

My tasks in the project:

- Responsible for the analysis of performance bottlenecks in the Anubex Framework
- Performace profiling of complex batch jobs Yourkit Profiler
- Suggest optimisation approaches
- Review partner of the external service provider
- Review and audit of the delivered sources

Technologies used in the project:

Technology / Frameworks	Tools	Databases	OS

Anubex, Anubex Dataturn, COBOL, Java	**Yourkit Profiler** IDEA, Docker, Docker Swarm	ORACLE 11	Linux ,VSE

Bilateral Initial Margin for OTC Trades

Lead Developer, Architect, DZ Bank Frankfurt am Main

April 2018 bis heute

For the determination of the bilateral initial margin (IM) to be exchanged, a completely new infrastructure is defined and implemented on the basis of the Standard Initial Margin Model (SIMM)TM. Based on the sensitivities calculated internally by the bank for the OTC transactions and their underlyings and other risk factors, the mapping to the likewise standardised CRIF format is carried out. By adapting the open source standard implementation of AcadiaSoft, a standardised IM determination of the market participants is to be achieved. The special challenges in this project lie not only in the mapping of existing internal processes and calculation methods to the ISDA SIMMTM model, but also in the enormous amounts of data that have to be processed with high performance.

The newly created processes are embedded in the existing risk systems or extend them with new functionalities.

In order to meet the changing requirements of the bank as a user of the system, large parts of the functionalities can be adapted by the end users by configuring dynamic sets of rules. This makes it possible to adapt the system to newer versions of the ISDA SimmTM standard as far as possible without code adaptations. It also enables what-if analyses before the introduction of a new version of the standard.

My tasks in the project:

- Responsible for the technical design and architecture of the subsystem
- Integration of the open source ISDA Simm calculation kernel
- Design and implementation of a rule engine
- **Implementation of automated, Excel-based, acceptance tests**
- Implementation of the technical components
- Support for the IM Dispute Tool developed in parallel

Technologies used in the project:

Technology / Frameworks	Tools	Databases	OS
Karaf OSGI, **AcadiaSoft Simm-Lib**, Finmath-lib, JBOSS Drools Decision Tables,Angular	IDEA, Powerdesigner	Sybase ASE	Linux, Solaris

Value at risk, aggregation of simulation results

Lead Developer, Architect, DZ Bank Frankfurt am Main

Juli 2015 bis April 2018

In view of the size of the derivatives portfolio and the significance of the same for the counterparty default risk of the bank, a bank-wide portfolio simulation was introduced to adequately map the risks in derivatives trading. The key figures relevant for risk assessment are executed daily as well as intraday in the form of automated, distributed calculations on a computing cluster. As a result, the data is sent to various systems in the bank and serves as a data basis for different processes there.

Access to the data is made possible via REST interfaces. Third-party systems also have the option of using various libraries provided by the project as an access layer. Clients in Python, C++, C# and Typescript currently exist via this route.

Within the scope of this project, the previous architecture will be migrated from JEE to an OSGI container architecture, among other things. The goal is to achieve even better scaling of the components as well as to create new possibilities to run invoices from alternative code versions in parallel.

The system is productive and is constantly being expanded with new products and methods.

My tasks in the project:

- Collaboration on the design and architecture of the overall system
- Refactoring of the existing components with regard to OSGI and services
- Implementation of individual technical components
- Migration to OSGI containers

Technologies used in the project:

Technology / Frameworks	Tools	Databases	OS
Karaf OSGI, Spring, Camel, Hibernate, JPA	IDEA, Powerdesigner	Sybase ASE	Linux, Solaris

Energy Trading Client, ComTrader

Review / Technological Audit, Deutsche Börse Eschborn

Juli 2016 - August 2016

Deutsche Börse AG offers its customers a highly available and distributed platform for energy trading. This platform is comparable to a normal exchange platform where market participants can offer their own local order books.

In addition to the actual backend, the exchange offers a proprietary Java FX trading client, which is made available to clients as an application. Due to the complexity of the traded products, the interface is also very complex and flexible. This complexity leads to repeatable load problems in times of heavy trading on the platform.

The specific objective of the review was to assess whether and where there was potential for performance improvements in the overall system. This type of external review is generally carried out for applications of the stock exchange and serves to ensure the quality of internal systems, which are mostly developed in-house.

My tasks in the project::

- Developing and conducting interviews with the technical stakeholders
- Review of the Architektur
- Static Code Analyse
- Performance Profiling on live test system
- Assessment of the architect with focus on "Fitness for Purpose" as well as "Fitness for Future".

Technologies used in the project:

Technology / Frameworks	Tools	OS
Java FX, Spring	IDEA, **Sonarqube**, **Yourkit Profiler**, **JArchitect**	Linux, Solaris, Windows

Index Calculation Engine

Review / Technological Audit, Deutsche Börse Eschborn

Juni 2016 - Juli 2016

The calculation and distribution of index data (DAX, SDAX, ...) is one of the main business areas of Deutsche Börse AG in Eschborn. The indices are calculated on the company's own systems and are subject to extremely high, non-functional requirements with regard to runtimes and error tolerance.

Specifically, in this project I conducted an evaluation of the new index calculation infrastructure currently under development. Given the criticality of the system, this type of external audit is generally carried out. In addition to the technical analysis carried out by me, the business requirements were also compared with the system.

My tasks in the project:

- Developing and conducting interviews with the technical stakeholders
- Review of the distributed architecture (messaging, disaster recovery)
- Review against Best Practice Approach GigaSpaces
- Static code analysis
- Evaluation of the architecture with a view to "Fitness for Purpose" as well as "Fitness for Future".

Technologies used in the project:

Technologie / Frameworks	Tools	OS
Java, JMS	Eclipse, **Sonarqube**, **JArchitect**	Linux, Windows

Development of a Social Casino App iOS and Android

Developer of the Spinarena.net mobile app for Android and iOS.

Januar 2020 bis heute

From 2018-2019, topics around mobile app development were added. Here, I focused on creating apps with as single a code base as possible. I now rely heavily on Flutter as a cross-platform environment. In addition to the non-public apps I have created a publicly available App.

The plan was to create a social casino app with the aim of giving ambitious slot players access to games from most relevant slot manufacturers in one app.

To date, almost 20 slot manufacturers have been connected to the app. Players are able to select slots in the app according to their criteria and then play them with virtual coins.

Since the application is to be available for iOS and Android, Flutter was used. Flutter enables the development of cross-platform applications and is now available2.2 in the version.

The app is productive both on the Web and as an App and is being actively developed.

Challenges of the platform

- Cross platform development with 1-2 developers and one code base for all platforms
- A wide variety of slot providers with a wide variety of technologies
- Attractive gamedesign and user interface

My tasks in the project:

- Solution Design App and Backend
- Implementation of the backend connection (Ously Backend, Firebase, Adjust, CustomerIO)
- Implementation of the entire app logic incl. layout, animations, etc.

Technologies used in the project:

Technologie / Frameworks	Tools	Databases	OS
Flutter, Dart, Java, Kotlin, Swift	Flutter 1, Flutter 2, Docker, Kubernetes, AWS	MariDB	Linux